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Q1. Research about Microsoft Azure Active Directory vs Microsoft Entra ID

Microsoft Azure Active Directory (Azure AD) was Microsoft's cloud-based identity and access management (IAM) service. It helped organizations manage users, applications, devices, authentication and access to cloud resources.

Microsoft Entra ID is the new name for Azure Active Directory. Microsoft renamed Azure AD to Microsoft Entra ID in 2023. The change is mainly a rebranding; the core identity and access management capabilities continue under the Entra product family.

Key points:

- Azure AD and Microsoft Entra ID refer to the same core cloud identity service.
- Entra ID provides authentication, authorization and access management.
- It supports features such as Single Sign-On (SSO), Multi-Factor Authentication (MFA) and Conditional Access.
- The name Azure AD may still appear in older documentation, tutorials and interfaces.

Q2. Basics of Linux Commands

Linux commands are instructions entered in a terminal to perform tasks such as managing files, directories, users and system processes. Some basic commands are:

Command	Purpose	Example
pwd	Shows the current directory	pwd
ls	Lists files and folders	ls
cd	Changes directory	cd Documents
mkdir	Creates a new directory	mkdir project
touch	Creates a new file	touch file.txt
cp	Copies files or folders	cp a.txt b.txt
mv	Moves or renames files	mv old.txt new.txt
rm	Removes files	rm file.txt
cat	Displays file contents	cat file.txt
clear	Clears the terminal	clear
man	Shows command manual/help	man ls

Q3. Types of Identity

In identity and access management, an identity represents a person, device, application or other entity that can be authenticated and given access to resources. Common types include:

- **User Identity:** Represents an individual user, such as an employee or student.

- **Device Identity:** Represents a computer, mobile phone, server or other device.
- **Application Identity:** Represents an application or software that needs to access resources.
- **Service/Workload Identity:** Represents a service or automated workload that accesses another service without a human user.
- **Guest Identity:** Represents an external user who is given controlled access to an organization's resources.

Q4. Difference between Users, Groups and Roles

User: A user is an individual identity that represents a person. A user can sign in and access resources according to the permissions assigned to them.

Group: A group is a collection of users or other identities. Groups make it easier to manage permissions for many users at once.

Role: A role is a set of permissions that defines what actions an identity can perform on a resource. Roles are commonly used for authorization and access control.

Feature	User	Group	Role
Meaning	Individual identity	Collection of identities	Set of permissions
Main purpose	Represents a person	Organizes identities	Controls allowed actions
Example	Alvin Shanty	BCA Students	Reader / Contributor / Owner